Internship Final Presentation

Yen Nguyen

FLOWMETRIC

Flowmetric is a pioneering company dedicated to enhancing environmental protection and economic resilience through advanced corporate water management solutions.

Mission: “Empowering water-intensive organizations with advanced technology to navigate the challenges of climate change and ensure a sustainable water future”.

Goal: To collaborate with water-intensive companies, enhancing their water efficiency and ensuring compliance with stringent environmental regulations and international standards.

Our Team

Wilder
CEO

Nam
COO

Yen
Intern

Responsibilities

01 Market Research

-Extensive market research on market landscape and the position of the company within it.
- Looking at other comparable companies, which provided valuable insights into the climate risk management space.

bluesign®

waterplan

Ceres

Southeast Asia Market

Textiles

Significant water challenges, including high water consumption, pollution, and regulatory pressures.
Demin producers and expand to other textile manufacturers.

Food & Beverage

Beverage production, brewing and bottling.
Food processors: maintaining high water quality and availability is critical for food processors.

Industrials

Rely heavily on water for cooling and other operational needs.
Cooling system
Water recycling

Agriculture

Agriculture in Southeast Asia consumes a substantial amount of water.
Irrigation
Manage water resources

Company	Compliance Mandate	Contact Name	Contact position	Link	rejected/rel	Note
ıınsung ectronics etnam				SamSungVietnam		Region: Bac Ninh, Thai nguyen
ıxconn ın Hai ecision ert Products etnam Co., d.	m/reports/CSR/EN/2022/Hontai_2022_Sustainability_Report_EN-			https://www.foxconn.com.vn/contact AnCham Vietnam-Intel Products Vietnam		fabrication, PCB fab, semiconductor assembly,cooling Wafer fabrication Cleaning and cooling(cleaning
ettel High chnology lustries rporation	https://viettel.com.vn/en/news-events/news/viettel-honored-as-a-sustainable-brand-			https://viettelhightech.vn/en/contact#map		high-tech electronics manufacturing
i Electronics etnam	https://www.lg.com/global/sustainability/circularity/					electronic components, display panels and semiconductor
sign etnam	https://www.renesas.com/us/en/company/sustainability			https://www.renesas.com/us/en/about		Subsidiary of Renesas Electronics Corporation
miconducto ietnam	Compliance: REACH, green partner			https://www.onsemi.com/company/about-onsemi/locations/dong-nai-province-vietnam		Dong Nai and Binh Duong Province

Responsibilities

Grant Research

02

-Focusing on identifying grants that support water sustainability and innovative companies working towards green initiatives.
-Looking at different grant opportunities and understanding their alignment with the company goal.

fundsforNGOs
Grants and Resources for Sustainability

United Nations Development Programme

P4G
Pioneering Green Partnerships, Investing in Impact

USAID
FROM THE AMERICAN PEOPLE

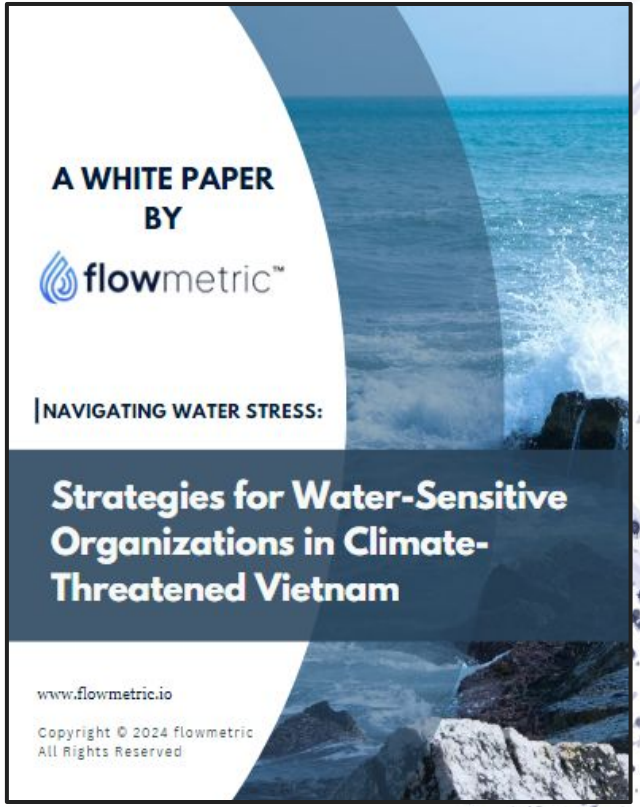
Grant Name	Administrator	Note	Criteria	Amount
Special Climate Change Fund (SCCF)	World Bank		Work with GEF (global environmental facility) Partner Agency to submit project proposals. Projects must be driven by the country, consistent with national	contact gra (request)
Green Climate Fund	World Bank	To receive money directly from the Green Climate Fund, entities need to be accredited as an "Accredited Entity" or AE. AEs can	Accredited Entities are responsible for monitoring and evaluating the projects they implement, as well as reporting their activities to GCF.	contact gra (request)
Green Climate Fund - Project Preparation Facility	World Bank		Funding proposals developed with the support of the Project Preparation Facility should be submitted to the GCF Board within two years of the approval of a PPF request.	to USD 1.5 m
Water Resilience Trust Fund under		ADB website for the	Support: Project Project preparation	contact gra (request)

Responsibilities

03

White Paper

-Working on a detailed report that explored the challenges and opportunities faced by water-sensitive organizations operating in Vietnam.
-Assist with the formatting of the document.



Responsibilities

04 Financial Impact

-Understanding of translating physical water risks and operational risks into measurable value.
-Involve analyzing how water scarcity and other water-related issues affect the financial performance.

Parameters
*Revenue:
*Fixed Assets
*Opex:
*Cost of Capital
Cost of water per M/3:
Total Annual Water Bill:
Total Annual Wastewater Bill:
Total Water Used (M3)
Total Wastwater Discharged
*Shadow Price of Water M/3

Scarcity Risk

Scenario	Annual Revenue ($)	# of D Outage
Low Scenario	300,000,000.00	
Medium Scenario	300,000,000.00	
High Scenario	300,000,000.00	

water-related damage

Flood Risk

Scenario	% of Asset Non-Performance	Fixed A
Low-Low	1%	650
Medium-Low	2%	650
High-Low	3%	650

Water inefficiency

Scenario	Water Tariff ($)	M/3 lost
Low loss scenario	0.51	
High loss scenario	0.51	

saving

Compliance Risk cost of compliance

Substance	Current Wastewater Treatment Cost ($)	Addition Treatme Infrastru
Lead	38,891.00	
Nickel	38,891.00	
Total Chromium	38,891.00	

Other Tasks

German Manufacture Research

Company Name	Industry Name	Location
Infineon Technologies AG	Semiconductor Manufacturing	Am Campeon 1- 85579 Neubiberg Germany
Elmos Semiconductor	Semiconductor Manufacturing	Heinrich-Hertz-S 44227 Dortmund Germany
Otto Graf	Plastic Manufacturing	Carl-Zeiss-Straße 79331 Teningen, Germany
Festo SE & Co. KG	Automation Machinery Manufacturing	Ruiter Str. 82, 73 Esslingen am Ne Germany
HYDAC Group	Machinery Manufacturing	Industriestraße, 6 Sulzbach/Saar, G
ProMinent Group	Industrial Machinery Manufacturing	Im Schuhmacher 5-11, 69123 Heid Germany
Bellmer	Machinery Manufacturing	Hauptstraße 37-4 75223 Niefern-Öschelbu Germany
KSB	Industrial Machinery Manufacturing	Johann-Klein-Str 67227 Frankenth Germany
Pepperl+FUCHS	Automation Machinery Manufacturing	Lilienthalstraße 1 68307 Mannhein Germany
Zchimmer & Schwars	Chemical Manufacturing	Chemnitztalstraß 09217 Burgstädt, Germany
FRIWO	Appliances, Electronics Manufacturing	Von-Liebig-Straf 48346 Ostbevern Germany

ASEAN state of water disclosure

- I had the opportunity to connect and network with organizations: ADB Bank, USAID
- Reaching out to these organizations and exploring potential collaborations and opportunities.

Contributions

Market and Grant Research

Provided actionable insights that helped shape strategic planning and market positioning.

Company	Compliance Mandate	Contact Name	Contact position	Link	Projected sale	Note	About	Date
Vinamilk				https://www.vinamilk.com.vn/vi/don-vi-thanh-		Food & Beverage	Dairy production, milk	
Vinh Hao				https://www.vinhhao.com.vn/lien-he		Food & Beverage	Mineral Water, bottling mineral water	
TH Group				https://thmilkfood1.valhalla10.stage.job		Food & Beverage	Dairy production, milk	
ACECOOK				https://acecookvietnam.vn/chi-nhanh/		Food & Beverage	Producer of instant noodles	
PepsiCo Vietnam				https://www.suntorypepsico.vn/		Food & Beverage	Beverage Drink	
Nestle Vietnam				https://www.nestle.com.vn/vi/aboutus/nestlevietnam		Food & Beverage	Coffee Market, MILO, MAGGI	
Tan Hiep Phat				https://www.thp.com.vn/ve-chung-toi/		Food & Beverage	Soft Drinks, energy drinks, and bottled water	

Priority	Company	Compliance Mandate	Contact Name	Contact position	Link	Projected sale	Note	About
	Delight Vietnam Industrial Company Limited	Bluesign			https://www.delightindustrial.com/en			Initial focus on producing greige textiles. Greige refers to unfinished textiles.
	MIRAE Fiber Joint Stock Company	Bluesign			http://miraejsc.com/en/			Padding, quilting, beddings, mattresses, and garment machinery
	ERE Saigon CO., LTD	Bluesign			http://erewebbing.com/			Produce webbing, elastic band, string
	Continental Textiles Co., LTD	Bluesign			https://continentaltextile.com.vn/contact			Hiigh-quality textile, fashion apparel to durable materials

Provided a comprehensive resource for the team to pursue funding opportunities.

Grant Name	Administrator	Note	Criteria	Amount	Deadline	Date Sent	Status	Grant Website	Description
FY2021 to FY2024 NOAA Broad Agency Announcement (BAA)	Department of Commerce		Request applications for special projects and programs associated with NOAA's strategic plan and mission goals, as well as to provide the general public with information and guidelines on how NOAA	contact grant (request)	30-Sep-24			HIGHERGOV: https://www.highergov.com/grant-opportunity/fy2021to-fy2024-noaa-broad-agency-announcement-baa-32926/#overview	This Broad Agency Announcement is a m encourage research, education and outrea projects, or sponsorships that are not addr NOAA's competitive discretionary pr
USAID Development Innovation Ventures (DIV) Program	Agency for International development		To be competitive under this solicitation, applications must be fully responsive to all directions within this APS document. Note that USAID may not award to an applicant unless the	$25,000 - $15,000,000	31-Oct-24			USAID Development Innovation Ventures (DIV) Program	USAID's goal of dramatically improving lives of the poor in developing countries, a ability to adhere to DIV's core princip provideexcellent value to the U.S. gove substantially contribute to these obj
P4G Call For Partnership Applications	Pioneering Green Partnerships, Investing in impact P4G		Applicants must provide services or products that contribute to poverty alleviation, gender equity and economic growth in one	USD 100,000 - 500,000 between 18-24 months	9/8/2024			P4G Call for Partnerships	P4G competitively selects partnerships early-stage climate entrepreneurs who in their seed or post-seed early growth sta to series A investor readiness

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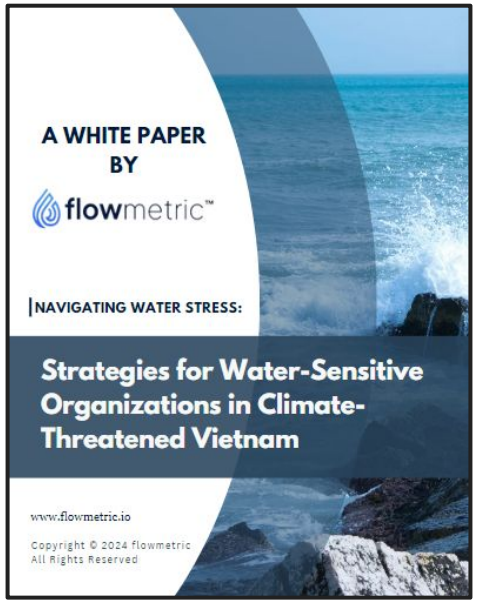
Special Climate Change Fund (SCCF)	World Bank		Work with GEF (global environmental facility) Partner Agency to submit project proposals. Projects must be driven by the country, consistent with national	contact grant (request)	none			Special Climate Change Fund (SCCF)	The Special Climate Change multilateral climate adaptati created at the 2001 Confe
Green Climate Fund	World Bank	To receive money directly from the Green Climate Fund, entities need to be accredited as an "Accredited Entity" or AE. AEs can	Accredited Entities are responsible for monitoring and evaluating the projects they implement, as well as reporting their activities to GCF	contact grant (request)	none			Green Climate Fund (GCF) | NDC Partnership	Set up in 2010 and aims to towards low-emission and c pathways by providing supp limit or reduce their greenhou to the impacts of
Green Climate Fund - Project Preparation Facility	World Bank		Funding proposals developed with the support of the Project Preparation Facility should be submitted to the GCF Board within two years of the approval of a PPF request.	to USD 1.5 millio	none			Green Climate Fund (GCF) - Project Preparation Facility	Set up in 2010 and aims to towards low-emission and c pathways by providing supp limit or reduce their greenhou to the impacts of
Water Resilience Trust Fund under the Water Financing	Asian Development Bank (ADB)	ADB website for the fund: https://www.adb.org/what-we-do/funds/wfpf-	Support: Direct Project (preparation and implementation) Program quality (policies,	contact grant (request)	contact grant			WFPF-WRTF	The fund aims to support c actions by government age private sector needed to stren and the

Contributions

White Paper

Enhanced the presentation of research findings, making them more accessible and engaging for stakeholders.



Contributions

Outreach

Engaged with organizations like the Vietnam USAID. Expanded professional relationships, strengthened the organization's network with key agency.

Yen Nguyen <yen.nguyen@flowmetric.io>
to Uyen

Dear Mrs. Uyen,

Thank you for your prompt response and for accommodating a quick phone call regarding the project information. We appreciate your time, especially given your busy training schedule.

Please let me know your availability and preferred time slots.

Thank you again for your assistance!

Best regards,

flowmetric

Yen Nguyen

yen.nguyen@flowmetric.io

https://www.flowmetric.io/

Yen Nguyen <yen.nguyen@flowmetric.io>
to Mbirnbaum, wilder.mccoy, nam.doan, Uyen

Dear Mr. Birnbaum,

I hope this message finds you well.

I would like to extend my sincere thanks to Mrs. Uyen for facilitating this connection.

Uyen mentioned that you will be in HCMC at the end of July. We would greatly appreciate the opportunity to meet with you in person. Please let us know your availability for a meeting.

We look forward to your response.

Best regards,

flowmetric

Yen Nguyen

yen.nguyen@flowmetric.io

https://www.flowmetric.io/

Achievements & Lessons

I have gained a deep understanding of the challenges that new businesses face, with those unique models like ours focused on water security and sustainability.

- Able to conduct comprehensive research and analysis.
- Deepened understanding of the climate risk management space, especially in Southeast Asia.
- Worked closely with team members on various projects, enhancing teamwork and communications skills.

Challenges

One of the most significant challenges I faced was adapting to a new industry and understanding its complexities. I initially felt overwhelmed by the technical terminology and intricate market dynamics. But investing time in reading industry reports and seeking guidance from my supervisor helps me learn and understand our company's structure.

Risk Typology
Water Quality Risk
Water Scarcity Risk
Flood Risk
Water Inefficiency
Compliance Risk
Repuational Risk (Loss)
Repuational Risk (Gain)

Water Tariff	M/3 of water Consumed
$ 0.51	146,350
$ 0.76	146,350
$ 1.01	146,350
$ 2.01	146,350

Water Tariff	M/3 of water Consumed
$ 0.51	146,350
$ 0.76	146,350
$ 1.01	146,350
$ 2.01	146,350

Recommendations

- Start projects early and take the time to explore host company and the related industry.
- More time you invest in understanding the company's operations and the industry's nuances, the more rewarding and impactful the experience will be.
- Opportunities for cross-departmental collaboration could provide a broader perspective.